Ajay Krishna Devulapally

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Professional Summary

Machine Learning Engineer and M.S. Computer Science graduate from Penn State with experience building reinforcement-learning systems, graph neural networks, computer-vision models, LLM/RAG prototypes, and evaluation-driven ML workflows. Strong background in Python, PyTorch, TensorFlow, model evaluation, benchmarking, failure analysis, data processing, and optimization, with growing experience in deployment-oriented AI services.

Education

The Pennsylvania State University — University Park, PA
M.S. in Computer Science, Thesis Track — *Aug 2024 – Aug 2026*

CVR College of Engineering — Hyderabad, India
B.Tech. in Computer Science and Engineering — *2019 – 2023*

Experience

Graduate Research Assistant — Aug 2024 – Aug 2026
Microsystems Design Lab, Penn State | SRC — *University Park, PA*

- Conducted SRC-supported research under **Prof. Vijaykrishnan Narayanan** on AI/ML-driven optimization for microsystems, chiplet placement, and combinatorial design problems.
- Developed RL-based 2.5D chiplet-placement algorithms using PPO, imitation learning, action masking, routing optimization, and thermal simulation for joint wirelength and thermal optimization.
- Built GNN-assisted freeze-mask optimization methods for oscillator-based Ising machines solving Max-Cut, using oscillator stability, flip dynamics, graph features, and dynamic node-freezing.
- Reduced OIM runtime by 33.8% on average across 7 external MQLib graphs versus the no-freeze baseline, with up to 54.1% runtime saved and 0.122% median Max-Cut loss.

AI Intern — Mar 2021 – Apr 2021
Smartknower — *Bangalore, India*

- Achieved 98.3% accuracy on MNIST by building ANN and transfer-learning image-classification models using TensorFlow and Keras.
- Implemented preprocessing, validation, evaluation, error analysis, and Streamlit demonstration workflows using Python, OpenCV, and Pandas.

Research

TWIRLeT-2.5D – First Author — 2026

*[**Submitted to TVLSI**] Ajaykrishna Devulapally, Varun Parekh, Anusha Devulapally, Mirco Scuilli, Srivatsa Rangchar Srinivasa, Vijaykrishnan Narayanan, "TWIRLeT-2.5D: Thermal Wirelength Integrated Reinforcement Learning for 2.5D Chiplet Placement."*

- Designed a graph-based RL environment modeling chiplet dimensions, power, connectivity, legal placements, orientation choices, thermal behavior, and routing-aware wirelength.
- Evaluated zero-shot inference on unseen configurations using temperature, wirelength, feasibility, runtime, and solution-quality metrics.

Projects

LLM/RAG Evaluation API | FastAPI, Hugging Face, FAISS, Docker — 2026 – In Progress

- Building a retrieval-augmented question-answering service over technical documents using document chunking, embeddings, FAISS vector search, grounded generation, and FastAPI endpoints.
- Designing evaluation workflows for retrieval accuracy, answer relevance, grounding, latency, and failure cases, with Dockerized execution and reproducible experiment configurations.

3D Human Pose Forecasting using SMPL & VPoser — 2025

- Built GRU- and Transformer-based PyTorch models to forecast 3D human-motion sequences from latent pose representations.
- Evaluated generated motion using MPJPE, PA-MPJPE, and MPVPE metrics for sequence-modeling and computer-vision analysis.

Skin Cancer Detection using Hybrid Deep Learning — 2023

- Achieved 90.625% classification accuracy using VGG16, ResNet, and CapsNet-style architectures for medical-image classification.
- Implemented preprocessing, transfer learning, model training, ensemble-style prediction, and quantitative evaluation workflows.

Skills

Programming / Data: Python, C++, C, SQL, NumPy, Pandas
ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
AI / ML: Reinforcement Learning, Imitation Learning, GNNs, Transformers, CNNs, RNNs, Computer Vision, NLP
ML Engineering: Model training, benchmarking, checkpointing, evaluation, error analysis, reproducibility, experiment design
Deployment / Emerging: FastAPI, Docker, FAISS, RAG evaluation, LangGraph, Git, Linux, OpenCV, Streamlit
Certificates: Artificial Intelligence – Smartknower (2021), Database Programming with SQL – Oracle (2022)